

ImageLab Progress Report Week 1

CMPS4191: Advanced Web Technologies

University of Belize

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<https://github.com/andreshungbz/imagelab>

IMAGELAB	Title	ImageLab Progress Report Week 1
	Course	CMPS4191 Advanced Web Technologies
	Semester	2026-1
	Due Date	Sep 9, 2026

Progress Summary

This first week involved laying the foundations of the project. Starting with the code from Quiz 2, the report-generation code was commented out, and the Go module and package references were changed to reflect the new GitHub repository.

Completed tasks included new database migrations (ImageLab consumer record, images table, image_variants table, and job failed_at column), new Go models (m_images.go, m_image_variants.go), new Go HTTP handlers and routes (h_images.go and refactored h_jobs.go), an initial draft of the image worker based on past code, and a frontend web application implementing the UI states using the event-emitter pattern.

In the API server, the CORS middleware and configuration code were reintroduced, as in the previous Advanced Databases final project, to account for the frontend web application starting separately from the server via the Live Server extension. The image handler validates the image's size and format, saves it to the server's disk, and returns a 202 Accepted response with placeholder values for a job.

In the frontend web application, the state variables, data service, application events, render functions, and handler setup were designed and implemented. Client-side validation was added via the form element and JavaScript, and the isSubmitting flag in the state object was used to disable the process image button. The HTML and CSS were also designed.

For the Week 1 check-in, none of its requirements were incomplete. Some parts are blocked, such as writing the image and job records, the worker generating image variants, and testing on the frontend regarding job status and results. Next, the worker will be fully implemented to generate image variants. Further testing will be conducted when these implementations become available.

One uncertainty to be resolved is how the consumer ID plays into the application. For now, it is configured as a server flag and never exposed to the client.